Emily Louie

emily.louie.ca emilyslouie@gmail.com github.com/emilyslouie linkedin.com/in/emilyslouie

EDUCATION

University of Waterloo – *Systems Design Engineering (Bachelors of Applied Science)* **2019 – 2024**

Courses: Data Structures and Algorithms (C++), Digital Systems (C) Cumulative GPA: 89% (3.95/4.0)

TECHNICAL SKILLS

Frontend: JavaScript/TypeScript, React, Redux, Gatsby, HTML, CSS/SCSS, GraphQL, REST

Backend: Java, Spring Boot, JUnit, C/C++, Golang, Python, Docker, Firebase, Azure, Neo4j, PostgreSQL, MongoDB

3D Graphics and Design: Unity, C#, Three.js, AFrame, Figma, Sketch, Adobe Creative Suite, Blender

EXPERIENCE

Figma – *Software Engineering Intern* **Jan – Apr 2022**

- Launched **3 new features** to enhance collaboration in FigJam which has 400,000+ weekly active users
- Consolidated numerous points of entry for users to increase productivity by **33%** using **TypeScript** and **React**
- Refined canvas snapping algorithms to improve user experience using **C++** with 100% testing coverage
- Built a system in TypeScript and React to display emoji reactions and custom CSS animations in real time for keywords typed in cursor chat for FigJam's first year anniversary, doubling usage from users

Unity – *Software Engineering Intern* **May – Aug 2021**

- Constructed and deployed **7+ advertising experiments** on Unity's ad network to 10+ billion users
- Generated **5% more revenue** by creating custom frontend layouts and animations in **TypeScript** and **React**
- Tracked metrics for 10+ A/B testing experiments by implementing numerous experiment policies in **Golang**
- Created a custom component library with 100% testing coverage that uses Storybook and Linaria

UW Blueprint – *Software Developer* **Aug 2020 – 2021**

- Created a full stack starter application with **MERN** and **Flask** for teams to develop software for nonprofits
- Implemented **GraphQL** and **REST** APIs with CRUD operations for **MongoDB** and **PostgreSQL**
- Designed a decision-making platform (arbitrium.web.app) that helped allocate **$2+ million** to local charities

Ontario Digital Service – *Software Engineering Intern* **Sep – Dec 2020**

- Developed and launched 3 new versions of Ontario's COVID-19 data visualizations (covid-19.ontario.ca/data) and informational microsites using **JavaScript** and **React**, averaging **450,000+ unique views** per month
- Reduced average build times by **52%** by optimizing data collection algorithms in the project's build process

Ontario Digital Service – *Product Design Intern* **Jan – Apr 2020**

- Collaborated with engineering teams to pioneer AODA/WCAG 2.0 compliant designs for Ontario's COVID-19 sites (covid-19.ontario.ca) which had **1.25+ million views** the month after launch
- Redesigned a web platform to help **5+ million users** renew their government identification cards annually

Microsoft – *Software Engineering and Product Design Intern* **Jul – Aug 2019**

- Developed the frontend of an internal inventory tracker used by **2,000+ employees** and reduces the request processing time by over **85%**, using **JavaScript**, **React** and **Azure**
- Presented the beta launch of the app to 800 executives and employees at the annual general meeting

PROJECTS

Six Degrees of Spotify – *Java, Spring Boot, Neo4j, Docker, Vaadin, JUnit* **/six-degrees**

- Built a full stack web application to explore that artists are connected by six or fewer song collaborations by maintaining a Neo4j database with **450,000** collaborations parsed from the Spotify API